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A Descriptive Subtitle If Needed

Anonymous submission

Abstract

LLM-based multi-agent systems (MAS) achieve strong performance on complex collaborative tasks, but their inter-agent communication opens an attack surface in which a few compromised agents propagate malicious behavior across the system. State-of-the-art defenses cast attacker detection as node classification on a multi-agent *utterance graph*, yet they compress each agent’s multi-turn dialogue into a single mean-pooled embedding before reasoning over the graph. We identify this temporal compression as a critical blind spot: because detection relies on an averaged representation, an attacker that starts benign and escalates its intent only gradually keeps its mean embedding close to that of honest agents, and thereby evades detection. Building on this insight, we make two contributions. First, we introduce an *escalation attack* that saturates the averaged feature with benign early turns, sharply degrading the detection rate of existing defenses while remaining effective at compromising the MAS. Second, we propose **T-GAT**, a temporal-aware detector that replaces mean-pooling with a temporal edge encoder fusing self-attention, temporal convolution, and cross-turn statistics (variance and step-wise change); by modeling *how* agents change rather than only *what* they say on average, T-GAT defends against both continuous and escalation attacks. Experiments on tool-attack scenarios across multiple LLM backbones and graph topologies show that the escalation attack lowers the baseline’s detection accuracy by [XX], while T-GAT restores detection to [XX] and reduces attack success rate by [XX] under both attack regimes.

Introduction

LLM-based multi-agent systems (MAS) now solve tasks that exceed the reach of any single agent, ranging from collaborative reasoning and debate to tool use and autonomous decision-making (Guo et al., 2024; Li et al., 2023; Zhuge et al., 2024). Their strength comes from inter-agent communication, but this same communication is a new attack surface: a few compromised agents can inject malicious content that propagates across the network and degrades the whole system rather than a single node (Yu et al., 2024). Such compromises stem from familiar vectors—direct and indirect prompt injection (Perez and Ribeiro, 2022; Greshake

et al., 2023), tool attacks (Zhan et al., 2024), and memory poisoning (Chen et al., 2024)—yet in a MAS their damage is amplified by propagation instead of being contained at the point of entry.

To defend MAS, recent work models the system as a graph and detects malicious agents from their communication. The state-of-the-art defense, G-Safeguard (Wang et al., 2025), constructs a multi-agent *utterance graph* whose nodes are agents and whose edges carry the messages exchanged between them, trains a graph neural network (GNN) to flag high-risk agents, and prunes their edges to halt contamination. Because the GNN is inductive, a single detector transfers across LLM backbones, topologies, and system scales without retraining. This detect-and-remediate paradigm is currently the most general and effective line of defense for MAS.

However, such detectors collapse the temporal dimension of multi-turn dialogue before they reason over the graph. Each edge accumulates an agent’s utterances across several rounds, but the detector compresses this sequence into one embedding by mean-pooling—or by keeping only the last turn—before message passing. Mean-pooling discards *how* an agent’s behavior evolves over the conversation, and we identify this as a critical blind spot: detection then depends on an agent’s average content rather than on its trajectory. An attacker can exploit this directly. If it behaves benignly in the early turns and escalates its malicious intent only gradually, the benign early turns dominate the average and pull the attacker’s mean embedding toward that of honest agents—saturating the very feature on which detection relies, so the attacker evades the detector while still steering the system.

We make two contributions that turn this blind spot into both a threat and a remedy. First, we introduce the *escalation attack*, a stealthy strategy in which a compromised agent starts indistinguishable from honest agents and ramps up its influence over successive turns (Figure ??). By saturating the mean-pooled feature with benign early turns, the escalation attack sharply lowers the detection rate of existing defenses while remaining effective at compromising the MAS. Second, we propose **T-GAT**, a temporal-aware detector that replaces mean-pooling with a temporal edge encoder. Rather than averaging an agent’s utterances, T-GAT extracts their temporal structure through self-attention over turns, a temporal convolution that captures local shifts, and explicit

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cross-turn statistics—variance and step-wise change—that directly expose escalation; a learned gate then fuses these temporal cues with message content before graph message passing.

The key insight behind T-GAT is that the discriminative signal is *how* an agent changes, not only *what* it says on average. An honest agent is consistent across turns, so its cross-turn variance is near zero; an escalating attacker is, by construction, inconsistent, so the statistics a mean obscures become its signature. Because T-GAT preserves these dynamics instead of discarding them, it is designed to detect escalation attacks that defeat mean-pooling while remaining effective against continuous attacks, whose strong per-turn signals existing detectors already capture. T-GAT keeps the inductive, drop-in design of graph-based detectors, so it inherits their transferability across backbones and topologies.

We evaluate on tool-attack scenarios built from InjecAgent (Zhan et al., 2024) across multiple LLM backbones and graph topologies. The escalation attack reduces the detection accuracy of G-Safeguard by [XX] and raises the attack success rate by [XX], while T-GAT recovers detection to [XX] and lowers the attack success rate by [XX] under both continuous and escalation attacks.

Contributions. We summarize our contributions as follows:

- We expose a temporal blind spot in state-of-the-art graph-based MAS defenses: by mean-pooling multi-turn dialogue, attacker detection depends on average content and ignores behavioral dynamics.
- We introduce the *escalation attack*, which exploits this blind spot by saturating the mean-pooled feature with benign early turns, evading detection while still compromising the MAS.
- We propose **T-GAT**, a temporal-aware detector that models cross-turn dynamics—self-attention, temporal convolution, and variance/step-change statistics—and defends against both escalation and continuous attacks while preserving inductive transferability.
- Experiments across multiple LLM backbones and graph topologies show that [summary of results once measured].

Related Work

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Method

Type your method here.

Experiments

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Conclusion

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Ethical Statement

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